

FEATURE BIBLE / VERSION 1.1

COMPETITIVE BIOME EXPANSION

Five visually distinct competitive arenas with living atmosphere, fixed competitive geometry, modular generated artwork, engine-authored animation, readable lose surfaces, scalable quality tiers, and zero biome-driven gameplay advantage.

DESERT	SNOWY MOUNTAIN	WHEAT FIELD	OCEAN	MUSHROOM FOREST
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Document Control

This revision preserves the complete Competitive Biome Expansion Specification and the complete Biome Art and Animation Production Standard. It distinguishes authored atmosphere from authoritative gameplay, then defines how static source art becomes modular, reusable, performant Godot animation.

DOCUMENT STATUS	CANDIDATE - approved for implementation planning and competitive validation
IMPLEMENTATION EVIDENCE	PLANNED - layered backgrounds and competitive presentation planning exist, but the reusable biome framework and five specified arenas are not implemented
PRODUCT SURFACES	Head-to-head, public online, ranked, private online, bot-supported competitive, and appropriate solo survival
CORE CONTRACT	Biomes change atmosphere, motion language, lose-state presentation, and ambience without changing the standardized arena, collision, physics, spawns, pickups, or outcome authority
SOURCES	Competitive Biome Expansion Specification and Biome Art and Animation Production Standard supplied by Don't Watch Media

Non-Negotiable Competitive Contract

BIOMES CHANGE THE FEELING OF THE MATCH - NEVER THE RULES THAT DECIDE IT.

The standardized arena dimensions, camera, player physics, spawn rules, pickup behavior, and fixed loss collision remain shared. Environmental motion is presentation-only, may desynchronize gracefully, and can be reduced or skipped for performance without invalidating the match.

Competitive Biome Expansion Specification

Feature Scope

Tap Against Me's head-to-head and online matches should support a growing collection of visually distinct competitive arenas. These biomes provide environmental personality, movement, and atmosphere without altering competitive balance.

The first five expanded competitive biomes are:

1. Desert
2. Snowy Mountain
3. Wheat Field
4. Ocean
5. Luminescent Mushroom Forest

These environments are intended for:

- Head-to-head matches
- Public online matchmaking
- Ranked competitive matches
- Private online matches
- Bot-supported competitive matches
- Solo competitive survival where appropriate

The biomes must preserve the standardized competitive play space. The background, ground appearance, ambient animation, and lose-state presentation may change, but the player physics, spawn rules, pickup behavior, camera framing, and arena dimensions must remain consistent unless a separate gameplay modifier is explicitly enabled.

1. Core Competitive Biome Rules

1.1 Visual Identity Without Competitive Advantage

Every biome should feel substantially different, but no biome should make a player harder to see or provide a gameplay advantage.

Environmental movement must never:

- Obscure a player for more than a brief moment.
- Visually resemble a collectible or hazard.
- Cover the primary gameplay area.
- Make one character skin significantly harder to see.
- Produce flashing or pulsing effects behind the players at high frequency.
- Change the player's collision, gravity, jump force, or movement.
- Cause network-dependent gameplay events.
- Introduce randomized physical obstacles in ranked matches.

Ambient animation exists to make the arena feel alive. It is not gameplay simulation.

1.2 Standardized Arena Structure

Every competitive biome should use the same structural layering:

1. Far sky
2. Far background
3. Midground scenery
4. Gameplay layer
5. Lose surface
6. Foreground decoration
7. Weather and atmospheric effects
8. HUD and reaction/emote layer

The players, pickups, and gameplay indicators must remain visually above the background and midground but below the HUD.

Foreground objects must remain near the outer edges or bottom of the screen. They should never hide a player during normal play.

1.3 Lose Surface

Each biome must clearly communicate the surface that ends the round.

Biome	Lose Surface
Desert	Sand-covered ground

Biome	Lose Surface
Snowy Mountain	Snow-covered mountain ledge or valley floor
Wheat Field	Soil and flattened wheat at ground level
Ocean	Water surface
Mushroom Forest	Glowing forest floor or fungal moss

The lose surface must occupy the same collision height in every standard competitive arena, even when the artwork makes the surface appear deeper, softer, or more irregular.

1.4 Day and Night Support

These competitive biomes should support TAM's approximately 120-second day-and-night cycle.

The cycle remains independent from gravity. Increasing gravity must not automatically darken the environment.

Each biome should define:

- Dawn color grading
- Day sky palette
- Sunset color grading
- Night sky palette
- Sun visibility
- Moon visibility
- Nighttime ambient lighting
- Biome-specific emissive elements
- Minimum visibility limits

Competitive visibility takes priority over physical realism. Night must never become so dark that players, pickups, the ground, or opponent indicators become difficult to read.

The host or authoritative match state should provide a shared biome time value so both players see approximately the same time of day.

2. Shared Biome Scene Architecture

Each biome should be built from a common reusable competitive biome scene.

Suggested structure:

```
CompetitiveBiome
+-- SkyLayer
|   +-- SkyGradient
|   +-- Sun
|   +-- Moon
|   +-- CloudSpawner
+-- FarBackground
+-- Midground
+-- GameplayBoundary
|   +-- LoseSurface
+-- Foreground
+-- AmbientEffects
+-- CreatureSpawner
+-- LightingController
+-- AudioController
+-- BiomeAnimationController
```

Each biome inherits or instantiates this structure and supplies its own assets and configuration.

Suggested biome resource fields:

```
biome_id
display_name
sky_palette
dawn_palette
sunset_palette
night_palette
ambient_light_day
ambient_light_night
cloud_density
wind_strength_visual
ground_scene
far_background_scene
midground_scene
foreground_scene
ambient_effect_scenes
creature_event_scenes
ambient_audio
lose_effect
```

The system should support enabling and disabling expensive background elements according to the player's performance settings.

3. Desert Biome

3.1 Concept

The desert biome should feel warm, open, and lightly windswept. It is not a violent sandstorm. The environment should appear calm enough for competitive readability while still showing continuous life through drifting sand, moving clouds, and subtle cactus motion.

The arena should communicate heat and distance without relying on aggressive screen distortion.

3.2 Visual Composition

Sky

The daytime sky uses a pale blue-to-warm-horizon gradient.

Clouds should be:

- Thin and widely separated.
- Slow-moving.
- Positioned primarily in the upper third of the screen.
- Slightly faded near the horizon.
- Large enough to read as distant scenery rather than gameplay objects.

At sunset, the horizon transitions through warm amber, orange, and muted purple.

At night, the desert can use a deep blue sky with strong moon visibility and a limited number of stars.

Far Background

The far background should contain:

- Layered dunes.
- Distant rock formations or mesas.
- Atmospheric haze near the horizon.
- Optional distant cactus silhouettes.

The far dunes should move only through subtle parallax. They should not animate like waves.

Midground

The midground can contain:

- Several cactus shapes.
- Small desert rocks.
- Dry shrubs.
- Sparse tufts of grass.
- Partially buried stones.

Cacti should remain behind the gameplay layer.

A cactus may bend by one or two pixels during stronger wind pulses, but it should not sway continuously like a tree.

Ground

The loose surface is a sand-covered floor with a clearly readable top boundary.

The sand can include:

- Small ripples.
- Occasional pebbles.
- Faint moving grains.
- Small puffs when a player lands.

The visual dune shape must not misrepresent the actual collision line.

3.3 Ambient Animation

Blowing Sand

Light sand should move horizontally across the lower portion of the arena.

The sand effect should use several animation layers:

- Very small grains moving quickly.
- Faint translucent streaks moving more slowly.
- Occasional low sand curls near the ground.

The effect should remain below the players whenever possible.

Recommended behavior:

- Baseline light drift at all times.
- Slightly stronger wind pulse every 8-18 seconds.
- Wind pulse lasting approximately 1.5-3 seconds.
- Randomized direction only when the biome is not meant to communicate a persistent wind direction.
- No full-screen opacity increase.

Clouds

Clouds move slowly across the sky on separate parallax paths.

Recommended cloud travel time:

- Approximately 35-70 seconds across the visible screen.
- Clouds should spawn outside the camera boundary.
- Cloud placement should avoid the center of the active play space when possible.

Heat Haze

A very subtle heat haze may appear immediately above the sand during daytime.

This effect should be:

- Restricted to a narrow lower-screen region.
- Disabled on low-performance settings.
- Disabled or heavily reduced during night.
- Mild enough that it does not distort player silhouettes.

3.4 Lose Animation

When a player touches the sand:

1. A small sand puff appears at the impact point.
2. The character briefly sinks several pixels into the sand.
3. A circular sand displacement moves outward.
4. The defeated character becomes visually subdued or transitions into the normal loss state.

This animation must not delay the authoritative loss result.

3.5 Ambient Audio

The desert soundscape can include:

- Light wind.
- Faint sand movement.
- Very occasional distant bird or desert-animal calls.
- Stronger wind texture during wind pulses.

Ambient sound should remain subordinate to gameplay effects, tap feedback, reactions, pickup sounds, and loss notifications.

4. Snowy Mountain Biome

4.1 Concept

The snowy mountain biome should feel cold, elevated, and crisp. The arena is positioned against a mountain range, with light snowfall, drifting snow, moving clouds, and subtle wind.

This biome should feel visually active without becoming a blizzard.

4.2 Visual Composition

Sky

The daytime sky uses cold blue and pale cyan tones.

At dawn and sunset, the snow should pick up soft pink, orange, and violet highlights.

At night, moonlight should create a cool blue tint across the snow while retaining strong player visibility.

Far Background

The far background should contain:

- Multiple mountain silhouettes.
- Snow-covered peaks.
- Atmospheric fog between mountain layers.
- Distant clouds partially crossing the peaks.

The mountain layers should use parallax to reinforce depth.

The farthest peaks should be lower contrast than the nearest background ridge.

Midground

The midground can contain:

- Snow-covered pine trees.
- Rocky ledges.
- Ice formations.
- Small piles of snow.
- Occasional distant cabins or lights, provided they remain subtle.

Pine trees may sway slightly during wind gusts.

Snow should occasionally fall from a branch when the branch moves, but this should happen in the background and not resemble a collectible.

Ground

The loose surface should be a snowy ledge, mountain shelf, or valley floor.

The snow surface must have a clearly defined top edge.

The ground can contain:

- Soft snow mounds.
- Patches of exposed rock.
- Small ice crystals.
- Wind-carved snow patterns.

4.3 Ambient Animation

Snowfall

Snowfall should consist of several depth layers:

- Tiny distant snowflakes moving slowly.
- Medium flakes moving at moderate speed.
- A small number of close flakes moving faster.

The central gameplay space should use lower snow density than the edges.

Snowflakes should not be the same color, size, or shape as pickups.

Recommended limits:

- No more than a small number of large foreground flakes at once.
- Reduced particle count during high-intensity gameplay.
- Snow density lowered automatically on performance mode.

Wind and Snow Drift

Wind pulses can push the snow at an angle for several seconds.

Near the ground, small lines of powder can move across the surface.

Recommended wind pulse:

- Every 10-22 seconds.
- Lasting 2-4 seconds.
- Smooth acceleration and deceleration.
- Background trees and snowfall respond to the same visual wind value.

Snow Shedding

A pine tree may occasionally release a small clump of snow.

Sequence:

1. Branch bends slightly.
2. Snow clump breaks apart.
3. Powder falls behind the gameplay layer.

4. Branch returns to its resting position.

This should be a low-frequency ambient event.

4.4 Lose Animation

When a player contacts the snow:

1. A soft powder burst appears.
2. The character creates a shallow indentation.
3. A small amount of snow falls back over the impact area.
4. The normal defeated-player state begins.

An alternate impact on exposed rock may produce a smaller powder effect, but all ground regions use the same loss collision.

4.5 Ambient Audio

The snowy mountain biome can use:

- High, soft wind.
- Faint snow movement.
- Occasional distant ice cracking.
- Sparse mountain bird calls.
- Soft snow-impact sounds during loss.

The environment should feel spacious and quiet rather than busy.

5. Wheat Field Biome

5.1 Concept

The wheat field should feel bright, open, peaceful, and constantly alive through wind. The primary environmental feature is the wheat itself, which moves in broad waves as gusts pass across the field.

The moving wheat must remain below the primary gameplay area.

5.2 Visual Composition

Sky

The daytime sky should be clear and inviting, with slow-moving clouds.

Sunset should be one of this biome's strongest visual phases, with golden light passing through the wheat.

At night, the wheat can appear dark gold or muted brown under moonlight while retaining edge highlights.

Far Background

The far background can include:

- Distant rolling hills.
- A farmhouse or barn silhouette.
- Windmills.
- Tree lines.
- Distant fences.
- Birds moving near the horizon.

These elements should remain subtle and should not make the environment feel crowded.

Midground

The midground consists of tall wheat arranged in multiple depth layers.

The wheat should not be rendered as one single animated surface. Separate layers create more convincing movement:

- Far wheat with small movement.
- Mid wheat with moderate movement.
- Near-ground wheat with stronger movement.
- Occasional taller stalks near the edges.

Ground

The lose surface is soil beneath a lower strip of wheat.

The top collision line should remain readable through:

- A narrow exposed soil strip.

- Flattened wheat near the gameplay boundary.
- A subtle outline or lighting separation.
- Consistent placement across all arena variations.

5.3 Ambient Animation

Wheat Sway

Wheat should move continuously at a low level.

The animation should not have every stalk moving in perfect synchronization. Use grouped phase offsets or a shader-driven wave.

The wheat movement includes:

- Small baseline sway.
- Larger rolling waves during wind gusts.
- Slight delay between far-left and far-right areas.
- Different sway strength by depth layer.

The wheat should bend rather than rotate rigidly from its center.

Wind Streaks

Visible wind streaks should occasionally move through the arena.

These streaks represent air movement rather than physical hazards.

Requirements:

- Thin and translucent.
- Primarily white, pale gold, or sky-tinted.
- Travel horizontally or at a slight angle.
- Fade in before entering the central play area.
- Fade out before reaching the opposite boundary.
- Never resemble projectiles.

Recommended frequency:

- Small wind streak group every 5-12 seconds.
- Larger gust every 15-30 seconds.
- Gust duration approximately 2-4 seconds.

During a larger gust:

- Wheat bends farther.
- Loose chaff or a few leaves move through the lower screen.
- Ambient wind audio briefly increases.
- Clouds may receive a very minor speed multiplier.

Distant Birds

A very small group of birds may cross the distant sky.

They should remain silhouettes and should not fly through the main player space.

5.4 Lose Animation

When a player reaches the ground:

1. Wheat stalks near the impact point bend outward.
2. A small dust or chaff effect appears.
3. The player settles into the wheat or onto the soil.
4. The stalks slowly return to position after the result is established.

The player should remain visible after losing and must not disappear completely behind the wheat.

5.5 Ambient Audio

The wheat field can include:

- Continuous soft rustling.
- Wind gusts.
- Occasional distant birds.
- Very faint rural environmental ambience.
- Chaff and grass movement during impact.

6. Ocean Biome

6.1 Concept

The ocean biome places the competitive arena above open water. Waves move continuously beneath the players, clouds pass in the sky, and dolphins occasionally jump from the ocean.

Unlike the other biomes, the water itself is the lose surface. Touching the water ends the player's round.

The water must therefore be visually readable as dangerous without appearing violent or hostile.

6.2 Visual Composition

Sky

The sky should feel open and expansive.

Possible background elements include:

- Soft clouds.
- Distant seabirds.
- A low horizon line.
- Subtle atmospheric haze.
- Sun or moon reflections.

The horizon must remain visually distinct from the active water surface.

Far Background

The far background may contain:

- Distant islands.
- Small rock formations.
- A lighthouse.
- A ship silhouette.
- Distant clouds touching the horizon.

These should be low-contrast and must not distract from the players.

Water Layers

The ocean should use multiple visual layers:

1. Distant horizon water.
2. Mid-distance wave layer.
3. Active surface wave layer.
4. Near-surface foam and splash layer.
5. Optional visible underwater gradient below the lose surface.

The actual loss collision remains a fixed horizontal boundary.

Wave peaks should not change the collision height. The animation can move above and below the collision line visually, but the average waterline must clearly communicate where loss occurs.

A subtle collision guide can be embedded into the water through:

- A brighter foam line.
- A slightly higher-contrast wave edge.
- A persistent thin highlight.
- Consistent waterline placement.

6.3 Wave Animation

Waves should be continuous and smooth.

The water should combine:

- Slow large wave movement.
- Faster small ripples.
- Occasional foam caps.
- Horizontal surface drift.
- Reflections that change with the day-and-night cycle.

The waves should not move so dramatically that players cannot judge their distance from the lose boundary.

Recommended visible wave displacement:

- Small relative to the character height.
- Collision remains fixed.
- Stronger visual waves can occur at screen edges rather than directly beneath the players.

6.4 Dolphin Asset and Behavior

The dolphin must be built as a separate reusable asset and scene rather than being baked into the ocean background.

Suggested structure:

```
DolphinEvent
+-- DolphinSprite
+-- AnimationPlayer
+-- EntrySplash
+-- ExitSplash
+-- WaterRipple
+-- AudioPlayer
```

The dolphin asset should support:

- Jumping from left to right.
- Jumping from right to left.
- Different horizontal distances.

- Slightly different jump heights.
- Optional single dolphin or paired dolphin events.
- Entry and exit splash positions.
- Day/night tinting.

Dolphin Event Sequence

1. A faint water ripple appears.
2. The dolphin breaks through the water.
3. An entry splash appears.
4. The dolphin follows a curved jump arc.
5. The dolphin rotates naturally through the jump.
6. It re-enters the water.
7. A larger exit splash appears.
8. Expanding ripples remain briefly.
9. The dolphin despawns below the water.

The dolphin remains behind the players and pickups.

Recommended event frequency:

- Approximately every 15-35 seconds.
- Do not trigger during the first few seconds of a match.
- Avoid triggering directly beneath a player near the lose surface.
- Avoid repeated events from the same location.
- Limit to one active dolphin event during standard matches.

The dolphin animation is cosmetic and does not interact with players.

6.5 Splash Effects

Splash particles must remain visually distinct from pickups.

Splash components can include:

- Water droplets.
- Foam.
- Circular ripples.
- A temporary water depression.
- A brief specular highlight.

Splash opacity and particle count should scale according to performance settings.

6.6 Player Lose Animation

When a player contacts the water:

1. The authoritative loss triggers immediately.
2. A player-specific splash appears at the contact point.
3. The player briefly enters the surface.
4. A circular ripple expands outward.
5. The player sinks slightly or transitions to the normal defeat display.
6. Small bubbles may rise for a brief period.

The loss splash should be larger and visually distinct from the dolphin splash so players immediately understand that a competitor has been eliminated.

The player's loss animation must not imply that the character is harmed. It should remain playful and suitable for TAM's tone.

6.7 Ambient Audio

The ocean biome can include:

- Continuous low waves.
- Small foam and water movement.
- Occasional distant seabirds.
- Dolphin splash and optional quiet dolphin vocalization.
- Larger splash when a player loses.

Ocean ambience should not mask tap, pickup, emote, opponent, or result sounds.

7. Luminescent Mushroom Forest Biome

7.1 Concept

The mushroom forest is a whimsical fantasy biome filled with oversized glowing mushrooms, soft spores, deep forest silhouettes, and a curious gremlin that occasionally moves between mushrooms.

The environment should feel mysterious and alive, but not threatening or horror-oriented.

The gremlin is an observer. It watches the competitors jump above it, hides when noticed, and runs between cover positions.

7.2 Visual Composition

Sky and Canopy

This biome may use a forest canopy rather than a fully open sky.

The upper background can include:

- Dark tree silhouettes.
- Gaps showing the sky.
- Hanging vines.
- Floating spores.
- Thin shafts of light.
- Distant glowing plants.

The sun and moon may still be visible through canopy openings during the shared day-and-night cycle.

At night, the mushroom glow becomes more visible, but the total arena brightness must remain safe for competitive readability.

Far Background

The far background should contain:

- Oversized mushroom silhouettes.
- Ancient tree trunks.
- Glowing fungal clusters.
- Soft fog.
- Small points of bioluminescent light.
- Distant moving spores.

Midground

The midground contains the primary glowing mushrooms.

Mushrooms should vary in:

- Height.
- Cap shape.

- Glow intensity.
- Stem width.
- Color family.
- Distance from the gameplay layer.

Large mushrooms should be placed toward the sides and behind the characters so they do not obscure gameplay.

Ground

The lose surface can consist of:

- Soft fungal moss.
- Small mushrooms.
- Glowing roots.
- Leaves.
- Dark soil.
- Bioluminescent patches.

The collision boundary should remain visible through a consistent glowing rim, moss edge, or illuminated ground strip.

7.3 Luminescent Glow Animation

The mushrooms should have a soft glow that fades in and out.

The entire biome must not pulse simultaneously.

Glow behavior should use:

- Independent phase offsets.
- Different pulse durations.
- Small intensity variation.
- Slow transitions.
- Limited bloom.

Recommended pulse duration:

- Approximately 3-7 seconds per cycle.
- Some mushrooms remain nearly steady.
- Smaller mushrooms may pulse slightly faster.
- Larger mushrooms should pulse slowly.

At night, the glow can become brighter relative to the environment without exceeding accessibility or bloom limits.

The glow must not flash.

7.4 Floating Spores

Small glowing spores may drift through the environment.

Requirements:

- Slow movement.
- Low particle density.
- Mostly behind the players.
- Different size and speed by depth.
- No resemblance to currency, apples, fruits, Hero Tokens, or other collectibles.

A few spores may respond to a gremlin running past by briefly scattering.

7.5 Gremlin Asset

The gremlin must be built as a separate reusable character asset.

The gremlin should look curious, mischievous, and non-hostile.

Suggested scene structure:

```
MushroomGremlin
+-- GremlinSprite
+-- AnimationPlayer
+-- CoverDetector
+-- PathController
+-- EyeDirectionController
+-- FootstepParticles
+-- AudioPlayer
```

Required animations:

- Hidden idle
- Peeking left
- Peeking right
- Watching upward
- Startled retreat
- Running
- Sliding behind cover
- Settling into hiding
- Optional blink
- Optional ear twitch or head tilt

The gremlin should never enter the gameplay layer or collide with players.

7.6 Gremlin Behavior Sequence

The gremlin event begins behind a large mushroom.

Phase 1: Initial Peek

1. The gremlin remains hidden.

2. Its eyes appear around the mushroom edge.
3. Its head slowly peeks outward.
4. It looks toward the nearest active player.
5. Its eyes track the characters as they move vertically.

The gremlin may switch its attention between players in head-to-head matches.

Phase 2: Observation

The gremlin watches for approximately 1.5-4 seconds.

Possible small actions:

- Blink.
- Tilt its head.
- Duck slightly when a player falls close to the ground.
- Look rapidly between both players.
- Hide briefly when a player performs a strong tap effect or reaction.

These reactions are cosmetic and occur locally.

Phase 3: Run Between Mushrooms

1. The gremlin checks the target mushroom.
2. It looks in the intended travel direction.
3. It dashes from behind the current mushroom.
4. It runs along a predefined background path.
5. Small leaves, spores, or dust move beneath its feet.
6. It reaches another mushroom.
7. It slides or dives behind it.
8. It peeks back out.
9. It resumes watching the match.

The gremlin should run beneath or behind the active player area.

Phase 4: Exit or Repeat

After reaching cover, the gremlin may:

- Remain hidden for the rest of the match.
- Peek one more time.
- Perform one additional run later.
- Retreat completely if the match ends.

Recommended event timing:

- First appearance no earlier than 8-15 seconds after match start.
- Approximately one major run every 20-40 seconds.

- No more than one active gremlin.
- Avoid running during the final result transition.
- Avoid crossing directly behind major UI elements.

7.7 Gremlin Spawn and Cover System

Mushrooms intended as gremlin cover should contain marker nodes.

Example:

```
GremlinCoverPoint
+-- HidePosition
+-- PeekLeftPosition
+-- PeekRightPosition
+-- EntryPosition
+-- ExitPosition
```

Each cover point should identify:

- Valid neighboring cover points.
- Whether the gremlin can peek left or right.
- Whether the point is visible within the current camera crop.
- Whether a path crosses the gameplay layer.
- The appropriate gremlin scale for the point's depth.

The system should choose only paths marked as competitive-safe.

The gremlin should not use freeform pathfinding. Predefined paths give the developer control over readability and animation quality.

7.8 Lose Animation

When a player contacts the fungal ground:

1. The moss compresses slightly.
2. A small cloud of spores rises.
3. Nearby tiny mushrooms wobble.
4. The ground glow briefly expands outward.
5. The character enters the normal loss state.

If the gremlin is visible, it may react by:

- Ducking behind its mushroom.
- Looking startled.
- Briefly peeking back out after the splash of spores.

The gremlin reaction is cosmetic and must not delay the result.

7.9 Ambient Audio

The mushroom biome can include:

- Soft forest ambience.
- Low magical hum.
- Faint mushroom glow tones.
- Quiet spore movement.
- Small gremlin footsteps.
- A subtle curious sound when the gremlin peeks.
- A quick rustle when it runs behind cover.

The gremlin should not make repetitive vocal sounds that become irritating over multiple matches.

8. Network and Determinism Requirements

8.1 Gameplay Authority

Biome animations must never determine competitive outcomes.

The authoritative match simulation controls:

- Player movement.
- Tap input processing.
- Gravity.
- Item collection.
- Debuffs and boosts.
- Collision with the lose surface.
- Elimination.
- Match results.

Biome systems only receive read-only match information needed for presentation.

8.2 Shared Cosmetic Seed

Each match can receive a cosmetic seed containing:

```
biome_id
environment_seed
match_start_timestamp
day_night_start_phase
```

The seed allows both clients to see approximately consistent:

- Cloud placement.
- Wind pulses.
- Dolphin events.
- Gremlin appearances.
- Snow gusts.
- Ambient timing.

Perfect frame-level synchronization is unnecessary because these events do not affect gameplay.

For spectators or recorded replays, the seed can reproduce the same general environmental event sequence.

8.3 Graceful Desynchronization

A cosmetic animation that is late, skipped, or different between clients must not:

- Change hit detection.
- Change the waterline.

- Change the ground collision.
- Trigger a loss.
- Block an input.
- pause the match.
- Cause the server to reject a valid gameplay state.

The client may skip an ambient event when device performance is low.

9. Competitive Readability Standards

Every biome should be tested with:

- Light player skins.
- Dark player skins.
- Bright particle effects.
- Transparent effects.
- Both players overlapping vertically.
- Day, sunset, night, and dawn lighting.
- Pickups active.
- Reactions or emotes active.
- Increased gravity.
- Small phone screens.
- Reduced brightness.
- Color-vision accessibility filters.

Recommended safeguards:

- Apply a subtle outline or rim light to both players.
- Maintain a minimum contrast ratio between characters and the background.
- Reduce background saturation near the central gameplay column.
- Keep the strongest environmental animation near the edges.
- Reduce bloom behind characters.
- Fade foreground objects when a player approaches them.
- Disable full-screen flashes.
- Offer reduced motion and reduced particle settings.

The central gameplay column should be visually calmer than the outer portions of the biome.

10. Performance Tiers

High Quality

- Full parallax layers.
- Complete weather effects.
- Dynamic glow.
- Full dolphin splashes.
- Gremlin eye tracking and secondary animation.
- Additional particles.
- Heat haze or advanced water shader.
- Higher cloud and spore counts.

Standard Quality

- Core parallax layers.
- Reduced particles.
- Simplified water reflections.
- Normal dolphin and gremlin events.
- Basic glow animations.
- Limited atmospheric distortion.

Performance Mode

- Static or minimally animated distant background.
- Lower particle counts.
- No heat haze.
- Simplified water shader.
- Reduced snowfall.
- Simplified wheat wave.
- Gremlin uses basic peek-and-run animations.
- Dolphin splash uses a short sprite animation.
- Reduced bloom.
- Fewer clouds.

Performance mode must preserve each biome's defining feature:

Biome	Feature That Must Remain
Desert	Light drifting sand

Biome	Feature That Must Remain
Snowy Mountain	Light snowfall
Wheat Field	Wheat movement
Ocean	Moving water and occasional dolphin
Mushroom Forest	Mushroom glow and gremlin appearance

11. Asset Requirements

Shared Assets

- Sun sprite
- Moon sprite
- Reusable cloud set
- Wind streak set
- Generic dust particles
- Generic splash particles
- Day/night gradient resources
- Player contrast or rim-light material
- Lose-surface shader interface

Desert Assets

- Multiple dune layers
- Mesa or rock formations
- Cactus set
- Dry shrub set
- Sand drift particles
- Sand impact animation
- Optional heat-haze shader

Snowy Mountain Assets

- Mountain layers
- Snow-covered pine trees
- Rock ledges
- Snowflake particle set
- Snow gust particles
- Branch snow-drop animation
- Snow impact animation

Wheat Field Assets

- Multiple wheat layers
- Individual wheat clumps
- Rolling hill background

- Farm silhouettes
- Wind streak particles
- Chaff particles
- Wheat impact animation

Ocean Assets

- Ocean surface tile or shader
- Multiple wave layers
- Foam animations
- Ripple animation
- Dolphin sprite sheet
- Dolphin jump animation
- Dolphin entry splash
- Dolphin exit splash
- Player loss splash
- Optional bubbles

Mushroom Forest Assets

- Large mushroom set
- Small mushroom set
- Glowing fungal clusters
- Forest silhouettes
- Spore particles
- Gremlin sprite sheet
- Gremlin cover markers
- Gremlin running particles
- Fungal ground impact animation

12. Recommended Development Order

Phase 1: Shared Framework

Build:

- Competitive biome base scene.
- Standard parallax structure.
- Lose-surface interface.
- Day/night hooks.
- Ambient event scheduler.
- Performance-quality settings.
- Shared environment seed support.

Phase 2: Wheat Field

The wheat field is a strong first implementation because it tests:

- Layered environmental animation.
- Shader or grouped sprite movement.
- Wind events.
- Readability against a moving background.
- Ground-impact reactions.

Phase 3: Desert and Snowy Mountain

These biomes expand the weather system through:

- Sand particles.
- Snow particles.
- Wind pulses.
- Background vegetation responses.
- Different lighting palettes.

Phase 4: Ocean

The ocean requires:

- Stable animated lose surface.
- Water shader or layered wave animation.
- Player splash loss effect.
- Reusable dolphin event scene.

- Ripple and foam systems.

Special attention must be given to ensuring animated waves never misrepresent the collision boundary.

Phase 5: Mushroom Forest

The mushroom forest is the most behaviorally complex environment because it introduces:

- Emissive animation.
- Multiple independent glow cycles.
- Gremlin state machine.
- Cover markers.
- Predefined movement paths.
- Match-aware observation behavior.

This biome should be implemented after the ambient event and reusable creature systems are stable.

13. Acceptance Criteria

A biome is ready for competitive use when:

- Both players remain clearly visible throughout the complete day/night cycle.
- The lose boundary is immediately understandable.
- Ambient animation does not affect physics.
- Ambient events operate without server communication after receiving the match seed.
- All major animation can be reduced or disabled through performance settings.
- No environmental object resembles a gameplay pickup.
- Players can correctly judge their height above the lose surface.
- The biome maintains stable performance on the minimum targeted Android device.
- The arena works in solo, private, bot, casual online, and ranked matches.
- The match-result sequence interrupts or safely concludes ambient creature events.
- Dolphin and gremlin assets operate as separate reusable scenes.
- The environment continues animating naturally without appearing repetitive during matches lasting two minutes or longer.

14. Final Biome Identities

The biomes should be immediately recognizable from their primary motion language:

- Desert: drifting sand, slow clouds, warm open distance.
- Snowy Mountain: falling snow, cold wind, layered mountain peaks.
- Wheat Field: broad rolling waves of wheat and visible wind streaks.
- Ocean: rhythmic waves, reflective water, dolphins breaking the surface.
- Mushroom Forest: breathing luminescent glow, floating spores, and a curious gremlin moving between cover.

Together, these arenas expand competitive TAM without fragmenting its gameplay rules. The biome changes what the match feels like, not how the match is won.

Biome Art and Animation Production Standard

This production standard is part of the feature contract. Generated artwork establishes the reusable visual components; Godot owns motion, timing, glow, particles, lighting, reactions, day-and-night transitions, and quality scaling.

STATIC SOURCE ART	TRANSFORMS	SHADERS	PARTICLES	SPRITE SHEETS
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1. Asset Creation Philosophy

The biome artwork should first be generated as polished, reusable static assets. Animation should then be implemented inside Godot using the most appropriate technical method.

Generated artwork provides the visual foundation. It should not contain baked-in motion that prevents the developer from controlling animation timing, intensity, performance, or interaction.

Each biome should therefore be separated into editable layers and independent assets, including:

- Sky backgrounds
- Distant scenery
- Midground scenery
- Foreground scenery
- Ground or lose surface
- Clouds
- Vegetation
- Creatures
- Environmental props
- Particle textures
- Lighting masks
- Glow masks
- Normal maps where useful
- Shader masks where useful

The developer should avoid using a single flattened biome image whenever an element needs to move independently.

2. Choosing the Correct Animation Method

Different environmental elements should use different animation techniques.

Sprite Sheets

Sprite sheets are best for elements with a defined illustrated motion sequence.

Use sprite sheets for:

- Dolphin jumps

- Dolphin water entry and exit
- Gremlin peeking
- Gremlin running
- Gremlin hiding
- Gremlin blinking
- Snow falling from a tree branch
- Player landing splashes
- Player sand impacts
- Mushroom wobble after an impact
- Small animal movement
- Specific environmental reaction animations

Sprite sheets should be used when the movement depends on changing the shape, pose, or silhouette of an illustrated subject.

Shaders

Shaders are best for continuous, repeating, or smoothly interpolated visual behavior.

Use shaders for:

- Mushroom luminescence
- Ocean surface movement
- Water reflections
- Heat haze
- Snow shimmer
- Sand shimmer
- Wind passing through wheat
- Soft cloud movement
- Day-and-night color transitions
- Fungal ground glow
- Gentle environmental distortion
- Subtle light shafts
- Animated emissive effects

Shaders should be preferred where generating individual animation frames would create unnecessary texture memory usage or visible looping.

Particles

Particle systems are best for numerous small, temporary objects.

Use particles for:

- Snowflakes
- Blowing sand

- Floating spores
- Water droplets
- Foam
- Dust
- Chaff
- Leaves
- Small sparks
- Sand impact puffs
- Snow impact powder
- Mushroom spores after landing

Skeletal or Transform Animation

Node transforms, bones, or simple procedural movement should be used when the original artwork can remain intact while its position, scale, rotation, or deformation changes.

Use these techniques for:

- Cactus bending
- Pine tree swaying
- Cloud movement
- Wheat layer movement
- Mushroom wobbling
- Background parallax
- Gremlin eye direction
- Sun and moon movement
- Distant bird travel
- Foreground object movement

3. Generated Asset Requirements

Generated biome artwork should be requested in a way that supports animation and implementation.

Whenever practical, assets should be generated with:

- Transparent backgrounds
- Clean silhouettes
- Consistent lighting direction
- Consistent perspective
- Consistent scale
- Minimal unwanted texture overlap
- Clear separation between movable parts

- Sufficient resolution for mobile displays
- Space around the object for animation
- No baked shadows that conflict with movement
- No text, UI elements, or accidental symbols

The generated art should match the established TAM visual style rather than using a different style for each biome.

For complex scenes, generate individual components instead of one complete flattened environment.

For example, the desert biome should use separate assets for:

- Sky
- Dunes
- Mesas
- Cacti
- Rocks
- Shrubs
- Clouds
- Sand particles
- Ground texture

This allows each element to receive its own parallax depth, movement, lighting, and optimization rules.

4. Shader-Driven Mushroom Glow

The luminescent mushrooms should primarily use shaders rather than sprite-sheet glow animations.

Each glowing mushroom should include:

- A base-color texture
- An emission or glow mask
- Optional noise texture
- Optional secondary color mask
- Individual phase offset
- Configurable glow intensity
- Configurable pulse speed
- Day/night intensity multiplier

The shader should animate emission smoothly over time.

The glow should not simply turn on and off. It should gradually breathe between low and high intensity.

Suggested conceptual behavior:

```
glow = base_intensity
      + sin(time * pulse_speed + phase_offset)
      * pulse_amount
```


A noise texture can slightly vary the glow across the mushroom cap so the entire surface does not brighten uniformly.

Each mushroom should receive a different phase offset so the biome does not pulse as one synchronized object.

Large mushrooms should generally pulse more slowly than small mushrooms.

At night:

- Emission becomes more visible.
- The base environment becomes darker.
- The mushrooms should not become overwhelmingly bright.
- Bloom must remain controlled.
- Player silhouettes must remain readable.

The glow shader should support a reduced-effects mode that replaces procedural variation with a simpler emission pulse.

5. Ocean Shader Requirements

The ocean should also rely primarily on shaders and layered textures rather than a large frame-by-frame water animation.

The shader can control:

- Horizontal wave distortion
- Vertical wave displacement
- Surface highlights
- Foam movement
- Reflection movement
- Day/night coloration
- Light shimmer
- Underwater gradient
- Subtle depth variation

The lose collision must remain fixed even when the visible waves move.

The shader must never move the apparent waterline so dramatically that players misjudge the danger boundary.

Dolphins and large splashes should remain sprite-based because they require clear, authored animation sequences.

6. Wheat Animation Requirements

The wheat field should use generated wheat artwork combined with shader-based or bone-based deformation.

The wheat should not require a large sprite sheet showing every stalk position.

Preferred methods include:

- Vertex deformation shader
- Layered wheat sprites with phase-offset rotation
- Bone deformation for large foreground clusters

- Shader mask controlling wind intensity across the field

A travelling wind value can move from one side of the field to the other, producing the appearance of a gust rolling across the wheat.

Foreground wheat may use stronger deformation while distant wheat uses smaller, slower movement.

7. Sprite-Sheet Standards

Sprite sheets should be planned before assets are generated.

Each animated character or creature should have a defined animation list, frame size, playback speed, and direction.

For example, the mushroom gremlin should have separate sprite-sheet sequences for:

- Hidden idle
- Peek left
- Peek right
- Look upward
- Blink
- Startled hide
- Run left
- Run right
- Slide behind mushroom
- Settle into cover

Frames should use:

- Identical canvas dimensions
- Consistent character alignment
- Stable scale
- Stable lighting
- Transparent backgrounds
- Clearly defined pivot points
- Adequate padding to prevent texture bleeding

Texture filtering and pixel-art settings should match TAM's final art direction.

8. Source Asset Preservation

The original generated images should be preserved separately from the game-ready exports.

Recommended asset structure:

```

art_source/
+-- biome_name/
|   +-- generated_originals/
|   +-- edited_layers/
|   +-- sprite_sheet_sources/
|   +-- masks/
|   +-- reference_images/

game_assets/
+-- biomes/
|   +-- biome_name/
|       +-- backgrounds/
|       +-- props/
|       +-- sprites/
|       +-- particles/
|       +-- shaders/
|       +-- materials/

```

Generated originals should not be overwritten when preparing textures for the game.

This preserves the ability to:

- Re-export at different resolutions
- Correct visual artifacts
- Change animation methods
- Create new biome variants
- Produce marketing artwork
- Adapt assets for future platforms

9. Performance Principle

The developer should not use a sprite sheet where a shader or transform animation can produce the same result more efficiently.

Similarly, a shader should not be used for a complex character performance that would be more controllable as an authored sprite animation.

The preferred order is:

1. Static generated artwork
2. Simple transform animation
3. Shader animation
4. Particle animation
5. Sprite-sheet animation
6. More complex simulation only when clearly necessary

This keeps biome animation modular, visually consistent, and practical for mobile hardware.

10. Final Production Rule

All competitive biomes should be treated as animated compositions built from generated visual assets.

The generated artwork establishes:

- Style
- Shape
- Texture
- Color
- Atmosphere
- Character design

Godot then provides:

- Movement
- Glow
- Weather
- Parallax
- Lighting
- Environmental reactions
- Creature behavior
- Day-and-night transitions
- Performance scaling

The goal is not to generate complete animated scenes externally. The goal is to generate high-quality art components that can be assembled, animated, optimized, and reused through the game engine.

Production Handoff

The source specification already defines the delivery order. This condensed implementation ledger makes each phase visible in the website tracker while preserving the complete biome chapters above as the authoritative feature reference.

Delivery Phases

1	Shared framework	Lock layer contracts, source/export folders, fixed lose-surface interface, day/night hooks, ambient scheduler, quality tiers, cosmetic seed, and animation-method rules.
2	Wheat Field	Prove separated generated art, shader or bone deformation, phase-offset wheat layers, travelling gust masks, readability, and impact reaction.
3	Desert + Snow	Add sand and snow particle families, shader shimmer and heat haze, transform-driven vegetation, authored loss sheets, palettes, and performance scaling.
4	Ocean	Protect a fixed water collision line while adding layered wave and reflection shaders, sprite-based loss splashes, and a reusable seeded dolphin sheet.
5	Mushroom Forest	Add masked emissive shaders, phase-offset glow, spores, reusable gremlin sheets and state machine, cover markers, authored paths, and reactions.
6	Competitive release	Audit source preservation, imports, pivots, texture settings, reduced effects, day/night, skins, pickups, accessibility, network, replay, memory, and Android gates.

Cross-Biome Release Evidence

- A single reusable competitive biome contract owns layer order, fixed lose-surface height, time-of-day inputs, ambient scheduling, quality scaling, seed reconstruction, and the source-to-export asset contract.
- Generated originals, edited layers, sprite-sheet sources, masks, and references remain preserved separately from optimized game-ready backgrounds, props, sprites, particles, shaders, and materials.
- Every animated element uses the least complex suitable method: transforms for intact objects, shaders for continuous interpolation, particles for numerous temporary elements, and sprite sheets for authored silhouette or pose changes.
- Every biome passes light/dark skin, pickup, particle, overlap, dawn/day/sunset/night, reduced brightness, small-phone, and color-vision readability tests.
- Environmental events remain cosmetic after seed receipt: no dolphin, gremlin, snow, sand, wave, wheat, spore, cloud, or light event can alter input, physics, collision, loss, or results.
- High, Standard, and Performance modes preserve each biome's defining motion language while meeting the minimum targeted Android frame-time, thermal, battery, and memory budgets.
- Two-device, reconnect, spectator, replay, bot, private, casual, ranked, and long-match tests prove graceful cosmetic drift and identical authoritative outcomes.
- Result transitions interrupt or safely conclude ambient creature events, and the separate dolphin and gremlin scenes can be reused without entering the gameplay layer.

TRACKER STATUS

PLANNED. Shared parallax art and day/night design provide useful foundations, but no reusable competitive biome scene or complete Desert, Snowy Mountain, Wheat Field, Ocean, or Luminescent Mushroom Forest arena is evidenced as implemented.